

Problem Set #2: Classical Mechanics (PH31207)

August 31, 2026

Generalized Coordinates and Constraints

1. Consider the following coordinate transformation:

$$x = r \sin \theta \cos \phi, \quad (1a)$$

$$y = r \sin \theta \sin \phi, \quad (1b)$$

$$z = r \cos \theta. \quad (1c)$$

When are these transformation equations not applicable.

2. Consider the constraint equation

$$(y^2 - x^2 - z) \dot{x} + (z - y^2 - xy) \dot{y} + x \dot{z} = 0.$$

Show that one has a holonomic constraint on the surface $z = x^2 + y^2 - xy$.

3. Consider the constraint equation

$$(yz - 2x + y) \dot{x} + (xz - 2y + x) \dot{y} + xy \dot{z} = 0.$$

Is this a holonomic or a nonholonomic constraint? Does this constraint relation satisfy the Frobenius integrability condition?

4. Using Frobenius integrability criterion, show that the constraint relations arising in rolling without slipping of a disk (on a 2D plane) are nonholonomic.
5. Consider the following classes of constraints: scleronomic, rheonomic, holonomic, nonholonomic, conservative, dissipative, bilateral, and unilateral. Now choose all the categories that apply to each of the following constraints.
 - (a) Simple pendulum with rigid support.
 - (b) Pendulum with variable length.
 - (c) A rigid spherical container of fixed radius R filled with gas.